MIT 1.125

AUTONOMOUS MARKETING & REVENUE INVESTIGATION AGENT

TERM PROJECT PROPOSAL

Liuyixin Shao

Problem and Audience

THE PROBLEM

- Dashboards show what changed, but explaining why still takes time.
- Identifying the driver requires SQL that connects traffic, behavior, orders, and refunds.
- Slow investigation delays decisions and action.

PRIMARY AUDIENCE

E-commerce Growth Manager

PROJECT GOAL

Give managers an AI analyst that investigates dashboard questions and explains KPI changes with traceable evidence.

Proposed Solution

Build a Streamlit business dashboard and a context-aware investigation agent.

1 DASHBOARD

- Shows core KPIs and trends
- Breaks results down by channel, device, and product
- Shares current filters as optional context

2 INVESTIGATION AGENT

- Answers natural-language questions
- Runs the SQL needed for each investigation
- Returns findings, charts, evidence, and limits

USER FLOW

See dashboard → **Ask a question** → **Agent investigates** → **Receive evidence**

Approach

DATA SOURCE: MAVEN FUZZY FACTORY

Public e-commerce dataset · 6 linked tables · 1.74M rows · 2012–2015

TECHNOLOGY STACK

DATABASE

SQLite

DASHBOARD

Streamlit + Plotly

MODEL

OpenAI GPT API

Analysis Tools

SQL + Python

AGENT INVESTIGATION LOOP

1	2	3	4
Interpret	**Plan**	**Analyze**	**Validate**
Question + context	Select evidence	Run SQL + Python	Check evidence

If evidence is insufficient, repeat steps 2–4

Output: findings, chart, evidence, limits

Link: https://mavenanalytics.io/data-playground/toy-store-e-commerce-database

Scope and Uncertainties

REALISTIC MVP

What we will deliver

1. **Working local prototype**

   A Streamlit dashboard and one investigation agent.

2. **Supported investigations**

   Revenue, acquisition, conversion, products, and refunds.

3. **Safe execution**

   Read-only SQL and Python with bounded runtime.

KEY UNCERTAINTIES

What could prevent success

Metric accuracy

Incorrect joins or KPI logic could produce misleading results.

Agent reliability

The agent may pursue weak leads or run too long.

Data coverage

No ad spend, refund reasons, customer feedback, or technical logs.

Plan and Success Criteria

FIVE-WEEK MILESTONE PLAN

WEEK	MILESTONE	DELIVERABLE
1	**Data**	Load SQLite; verify joins and core metrics
2	**Dashboard**	Build KPI trends, breakdowns, and filters
3	**Analysis**	Build tools; define 16 gold cases
4	**Agent**	Log trajectories; run 8 development cases
5	**Evaluation**	Run 8 held-out cases; score results

EVALUATION PLAN

16
TEST CASES

8 dev + 8 held-out
Harness logs full trajectories.

SUCCESS CRITERIA

Find
Finds required evidence and the key driver.

Investigate
Uses relevant, correct, non-redundant steps.

Explain
Returns findings, evidence, and limits.